

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
S Susmitha	192571068	ITA0510	Computer Vision

38. Face Detection using Opencv.

AIM:

Face Detection.

PROGRAM:

```
import cv2

# Load the Haar Cascade for face detection
face_cascade = cv2.CascadeClassifier(
    cv2.data.harcascades + "haarcascade_frontalface_default.xml"
)

# Open the webcam
cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Error: Could not open webcam!")
    exit()

while True:
    ret, frame = cap.read()

    if not ret:
        print("Error: Could not read frame!")
        break

    # Convert to grayscale
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Detect faces
    faces = face_cascade.detectMultiScale(
        gray,
        scaleFactor=1.1,
        minNeighbors=5,
        minSize=(30, 30)
    )
```

```
# Draw rectangles around detected faces
for (x, y, w, h) in faces:
    cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 0), 2)

# Display the result
cv2.imshow("Face Recognition", frame)

# Press 'q' to quit
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()
```

OUTPUT:

It will access the cam and detect the face.

RESULT:

Thus the program is executed successfully.